# United International University (UIU)
## Dept. of Computer Science & Engineering (CSE)
## Final-Term Exam Trimester: Spring 2025
**Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers**
**Total Marks: 40 Duration: 2 hours**

**Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.**

**Question 1: Answer all the questions. (10 Marks)** **[CO3]**

| Global Descriptor Table | |
|---|---|
| Index | Address |
| 045H | Base: 23308810, Limit: 18BD7H, Access: 54H, G = 1 |
| 1A3H | Base: 4FE4DBB8, Limit: 2345AH, Access: 1FH, G = 0 |
| 2F6H | Base: 866125A3, Limit: 1132FH, Access: 8AH, G = 1 |
| 4D1H | Base: A2B2ACFF, Limit: 256A9H, Access: 27H, G = 0 |
| 6B4H | Base: C61310BD, Limit: 10AEAH, Access: 53H, G = 1 |
| A29H | Base: E4DF02F6, Limit: C5072H, Access: 49H, G = 0 |

An **80386 microprocessor** is executing a segment in protected mode and a portion of its descriptor table is given above. If the current physical address produced by this processor is **92A15BF8H**, then determine the followings:

| | | |
|---|---|---|
| a. | Find out the **descriptor (index number)** of which this physical address belongs to. | **[4]** |
| b. | For a physical address of **867003A0**, determine the offset. | **[2]** |
| c. | Has the segment been accessed? | **[1]** |
| d. | Segment type (**CS/DS/SS/ES**). | **[2]** |
| e. | Is the descriptor defined or undefined? | **[1]** |

**Question 2: Answer all the questions. (10 Marks)** **[CO5]**

| | | |
|---|---|---|
| a. | In a microcontroller where a **10-bit digital-to-analog (DAC)** resolution is configured, the "**analogWrite(x)**" instruction is invoked for digital-to-analog conversion programming, resulting in a calculated duty cycle of **60%**. Given a system voltage of **5V**, what is the value of **x**? Also, **Draw the duty cycle diagram.** | **[3+1]** |
| b. | In a microcontroller-based system a master device, **Master (index:5H)** sends **2 byte** data (**char 'Micro'**) to the **slave (index: A3H)** using **I2C**. Draw the corresponding sequence diagram. **ASCII** codes for **'A'** and **'a'** are **65 and 97 respectively**. | **[4]** |
| c. | Suppose you are designing a **real-time missile guidance system** where split-second decisions are critical. Since the system must fetch instructions and process sensor data simultaneously without delay, speed is the top priority. With no limitations on cost or design complexity. The system can be built using either **Harvard Architecture or Von Neumann Architecture**. **Which architecture would be more suitable** and **why?** | **[2]** |

## Question 3: Answer all the questions. (10 Marks) [CO3]

a. Suppose there are **nine** instructions loaded into the Instruction Memory (within the Code Segment of 8086). The BIU of the 8086 will fetch these instructions for execution by the EU. **[3+3]**

**When the one instruction is being executed, two instructions are fetched and saved in the instruction queue.** Draw the corresponding BIU and EU cycles for the following cases.

I. The **3rd** instruction is a '**JUMP <6th instruction address>'**, the **4th** is a **'JUMP <8th instruction address>'** and the **7th** instruction is a **'MOV <address>'** instruction.

II. The **2nd** instruction is a '**JUMP <6th instruction address>'**, the **4th** is a **'JUMP <8th instruction address>'** and the **5th** instruction is a **'JUMP <3rd instruction address>'** instruction.

b. Draw the timing diagram for **Memory Read** operation of microprocessor "808x" showing the activities of Address-Status and Address-Data pins ($A19$-$A16/S6$-$S3$, $AD15$-$AD0$), $\overline{ALE}$, $\overline{DEN}$, $RD$, $DT/\overline{R}$, $\overline{M}/IO$, $\overline{WR}$ in each clock cycle. **[Observe the pins carefully]** **[4]**

## Question 4: Answer all the questions. (10 Marks) [CO4]

### RAM Content

| Address | Content | | Address | Content |
|---|---|---|---|---|
| 0H | | | 8H | |
| 1H | | | 9H | |
| 2H | | | AH | |
| 3H | | | BH | 0000 0101 |
| 4H | | | CH | 0001 0010 |
| 5H | | | DH | 0000 0001 |
| 6H | | | EH | 0000 0110 |
| 7H | | | FH | 0000 0011 |

### Opcode Table

| Mnemonic | Opcode |
|---|---|
| LDA | 0011 |
| ADD | 0001 |
| SUB | 0010 |
| OUT | 1101 |
| HLT | 1011 |

a. Write the **assembly code** for the following expression using **RAM content**. **[4]**

$$(3 + 1) \times 2 - 5$$

b. Fill-up the **RAM content table** with the instructions **machine code (Start from 0H)**. Use the opcodes given in the **opcode table**. **[3]**

c. Write the control words for Execution T-states (**T4, T5, T6**) to perform the **SUB instruction** using the controller sequence below. **[3]**

**CON** = $\overline{L}_I \overline{E}_I \overline{L}_A E_A \; E_p C_p L_M \overline{C}_E \; E_U S_U \overline{L}_O \overline{L}_B$ **[Observe the pins carefully]**